

The empirical recurrence for OEIS A269212, recovered from its tail

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Abstract

OEIS A269212 records “Empirical recurrence of order 42” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 214$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 44$. Its last coefficient is $c_{42} = -18249984 \neq 0$, so no further tail satisfies anything shorter; any order-42 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A269212 is “Number of $n \times 6$ 0..3 arrays with some element plus some horizontally, diagonally or antidiagonally adjacent neighbor totalling three exactly once.”. It has offset 1 and begins

1620, 360960, 70371048, 11985002256, 1910809190712, 292321215814512, 43468685827935816, 6331185189881558

The entry states:

Empirical recurrence of order 42 (see link above)

As of the “Last modified” line on the live entry (Feb 20 2016 11:28:16, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 214$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 214$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 42: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 42.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 428$ exact terms from $n = 2$ on, it returns order 42 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{42} c_i a(n-i),$$

c_1	328	c_{12}	337368544562593	c_{23}	3526772441744847394	c_{34}	−380030670631320
c_2	−36148	c_{13}	5933978699018136	c_{24}	−5744941204088426004	c_{35}	−681893660222512
c_3	1496854	c_{14}	−25380047778921801	c_{25}	−696359991304833804	c_{36}	35679798447004
c_4	−16592528	c_{15}	1503877510430024	c_{26}	4177526174778007587	c_{37}	12908198926272
c_5	−337427348	c_{16}	204377175273958799	c_{27}	−874427242915861010	c_{38}	−752611059712
c_6	7375607273	c_{17}	−360642114401056904	c_{28}	−1502268132727418865	c_{39}	−106297015872
c_7	−1052387678	c_{18}	−468440756929648493	c_{29}	600534645507554284	c_{40}	6217608192
c_8	−813108690446	c_{19}	1892193208214328166	c_{30}	252106818416592932	c_{41}	309224448
c_9	4410299596484	c_{20}	−482363509280326928	c_{31}	−147604348046802852	c_{42}	−18249984
c_{10}	26214904063798	c_{21}	−3942862366991878776	c_{32}	−14580425317202680		
c_{11}	−300831810190434	c_{22}	3583951997657359612	c_{33}	15812903799668816		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 44$, and it is the recurrence OEIS A269212 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{42} - c_1 x^{42-1} - \dots - c_{42}$. Its constant term is $-c_{42} = 18249984$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 42 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 42, so $\deg q = 42$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 42, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 11 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 42, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -18249984 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-42 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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